MCS4302 Mobile Computing — Week 5 Lab

Measurement Sheet: State Restoration, Background Jobs and Tail-Energy Model

Student index: 24440639	**Emulator API level:** 37
Repository: github.com/lucky1993-hub/MCS4302-Week4-Lab	**Part 2:** Restoration implemented; item persisted. Draft-restoration outcome not confirmed.

Part 2 — State restoration

Observed/known: restoration scope and Restorable state were implemented; the item was observed to persist. Draft-text restoration was still being investigated, so its final outcome is marked not confirmed rather than assumed.

Part 4 — Background job measurements

Test	Scheduled	Started	Ran?	Extra delay	Within watch?
J1 — Run 1	22:46:09.769	22:48:11.363	Yes	61.6 s	Yes*
J2 — Doze	22:50:14.347	22:51:39.209	Yes	24.9 s	Yes*
J3 — Rare	22:52:55.849	22:54:21.100	Yes	25.3 s	Yes
J4 — Restricted	22:55:13.779	22:56:38.686	Yes	24.9 s	Yes

*Within five minutes of scheduling based on the supplied timestamps. J2 completed, but whether IDLE_MAINTENANCE was reached or fallback used was not confirmed.

Part 5 — Tail-energy model

Calculation	Result
Active: 1.2 W × 1 s = 1.2 J; tail: 0.6 W × 10 s = 6.0 J	Per wake-up = 7.2 J
Separate sync: n = 60; 60 × 7.2 J	432 J/hour = 0.12 Wh
Batched sync: n = 4; 4 × 7.2 J	28.8 J/hour = 0.008 Wh
Tail fraction: 6.0 / 7.2 × 100%	83.33%
Modelled reduction: (432 − 28.8) / 432 × 100%	93.33%

Interpretation

The model assigns 83.33% of each wake-up's energy to the radio tail, so reducing separate wake-ups can reduce modelled energy: 432 J/hour for 60 wake-ups versus 28.8 J/hour for four. This 93.33% reduction is illustrative, not a battery-life prediction, because transfer durations and radio behavior vary. WorkManager's deferral can suit non-urgent sync, while time-sensitive chat notifications should use push messaging; the manually triggered J2 test does not measure how often maintenance windows occur.

Data-quality note: The final Part 2 draft-restoration outcome and the J2 Doze-state/fallback outcome were not confirmed. These remain marked not confirmed rather than assumed. The actual job runs are reported as observed.